CORE PURE / COMPLEX NUMBERS

# Loci and Regions in the Argand Diagram

Worked solutions 10 questions · 87 marks

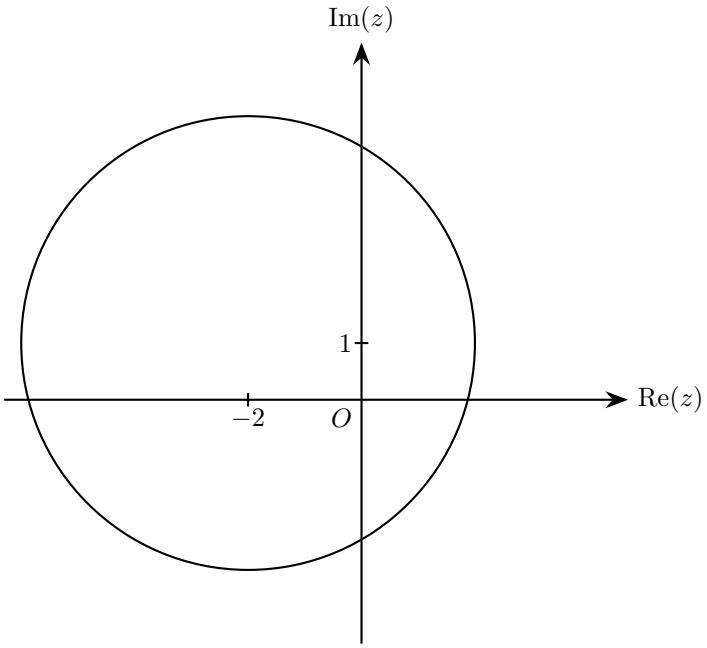
EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

**Instructions**

Cambridge International: this topic belongs to A Level Mathematics (9709), Paper 3 and is assumed knowledge for Further Mathematics 9231.

## Questions


| | | |
|---|---|---|
| Question 1 | *6 marks* | 2 |
| Question 2 | *6 marks* | 4 |
| Question 3 | *10 marks* | 7 |
| Question 4 | *11 marks* | 11 |
| Question 5 | *11 marks* | 14 |
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| Question 10 | *7 marks* | 31 |




**1.** The Argand diagram shows a circle of radius 4. The centre of the circle is the point which represents the complex number $-2+i$.

**(a)** Use set notation to define the locus of complex numbers, $z$, represented by points which lie on the circle. **[2]**

The locus $L$ is defined by

$$L = \left\{ z \in \mathbb{C} \;\middle|\; |z-(2+i)| = |z+1-3i| \right\}$$

**(b)** On a copy of the Argand diagram, sketch and label the locus $L$. **[2]**

You are given that the locus

$$\left\{ z \in \mathbb{C} \;\middle|\; \arg(z+1-2i) = \frac{2\pi}{3},\ \mathrm{Im}(z) = 5 \right\}$$

contains only one number.

**(c)** Find this number. **[2]**

**Solution**

**(a)** The circle has centre $-2+i$ and radius 4.

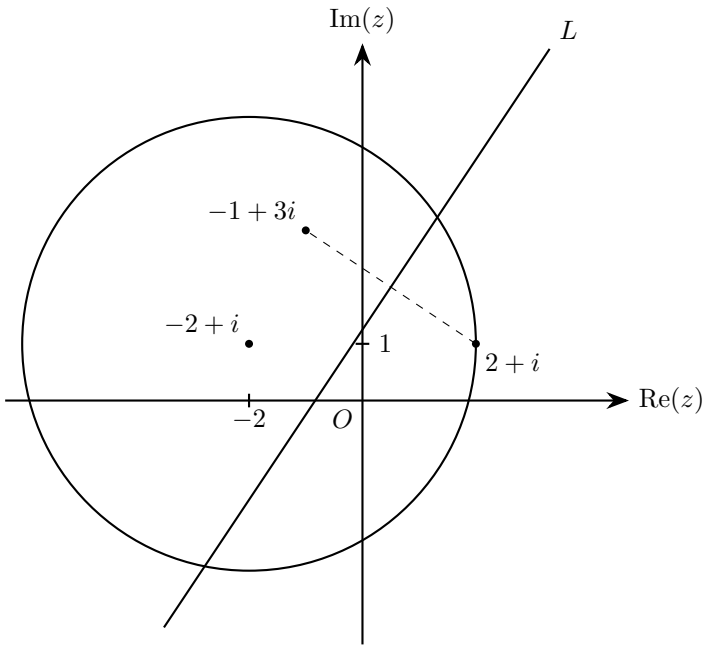
So the locus is the set of complex numbers whose distance from $-2+i$ is 4:

$$\{z \in \mathbb{C} \mid |z-(-2+i)| = 4\} = \{z \in \mathbb{C} \mid |z+2-i| = 4\}$$

**(b)** Since $|z+1-3i| = |z-(-1+3i)|$, $L$ is the set of points equidistant from $2+i$ and $-1+3i$, so it is the perpendicular bisector of the line segment joining those two points.

It passes through the midpoint of that segment:

$$\left(\frac{2+(-1)}{2}, \frac{1+3}{2}\right) = \left(\frac{1}{2}, 2\right)$$




**(c)** Since $\mathrm{Im}(z) = 5$, write

$$z = x + 5i$$

Then

$$z + 1 - 2i = (x + 1) + 3i$$

We are given

$$\arg(z + 1 - 2i) = \frac{2\pi}{3}$$

So $(x + 1) + 3i$ lies in the second quadrant, with $x + 1 < 0$. Therefore

$$\tan\frac{2\pi}{3} = \frac{3}{x+1}$$

Using $\tan\frac{2\pi}{3} = -\sqrt{3}$:

$$\frac{3}{x+1} = -\sqrt{3}$$

Hence

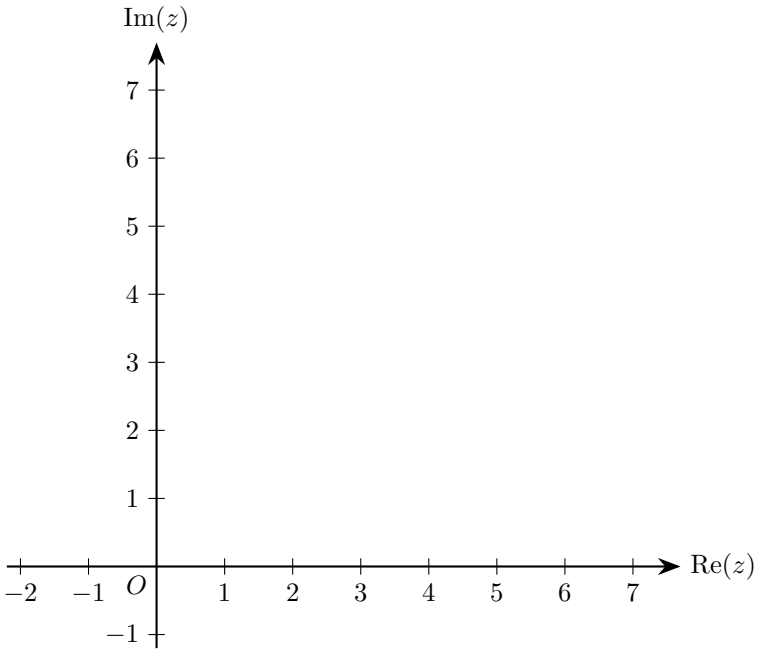
$$x + 1 = -\sqrt{3}$$

so

$$x = -1 - \sqrt{3}$$

Therefore the required number is

$$z = -1 - \sqrt{3} + 5i$$




**2.** **(a)** Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (2 + 4i)| = 3$$ **[2]**

**(b)** Sketch, also on the Argand diagram, the locus of points satisfying the equation

$$\arg(z - 1) = \frac{\pi}{4}$$ **[1]**

**(c)** For the complex number $w$, find the maximum value of $|w|$ such that

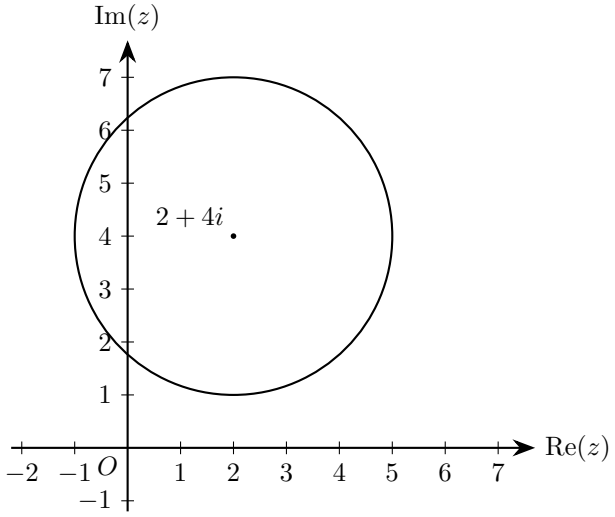
$$|w - (2 + 4i)| \leq 3 \qquad \text{and} \qquad 0 \leq \arg(w - 1) \leq \frac{\pi}{4}$$ **[3]**

**Solution**

**(a)** The locus

$$|z - (2 + 4i)| = 3$$

is a circle with centre at the point representing $2 + 4i$ and radius 3.



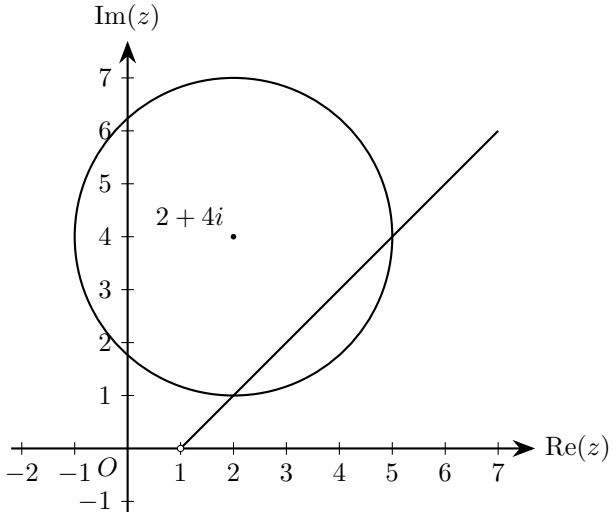

**(b)** The locus

$$\arg(z - 1) = \frac{\pi}{4}$$

is the half-line starting at the point representing 1 on the real axis, making an angle of $\frac{\pi}{4}$ with the positive real axis.

In Cartesian form this is

$$y = x - 1, \qquad x > 1$$




**(c)** Let

$$w = x + iy$$

The condition

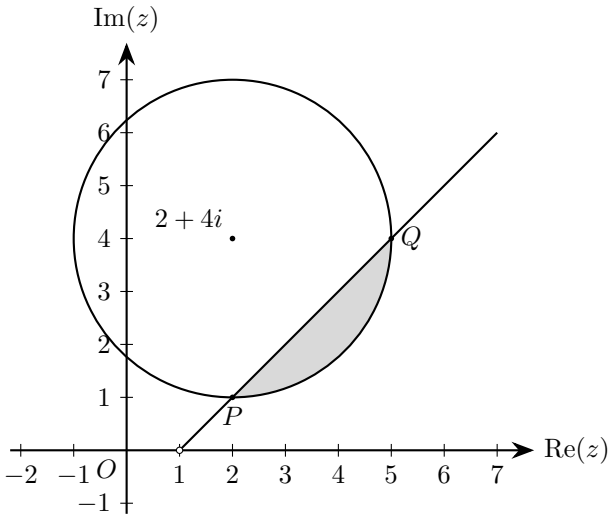
$$|w - (2 + 4i)| \leq 3$$

means that $w$ lies inside or on the circle from part (a).

Also, the condition

$$0 \leq \arg(w-1) \leq \frac{\pi}{4}$$

means that $w$ lies between the half-line from 1 along the positive real axis and the half-line from part (b).




Since the circle has lowest point $(2, 1)$, every point in the circle has $y \geq 1$, so the condition $\arg(w-1) \geq 0$ is automatically satisfied.

Therefore the required region is the part of the disc lying on or below the line

$$y = x - 1$$

First find where this line meets the circle:

$$\begin{aligned}
(x-2)^2 + (y-4)^2 &= 9 \\
(x-2)^2 + (x-1-4)^2 &= 9 \\
(x-2)^2 + (x-5)^2 &= 9 \\
x^2 - 4x + 4 + x^2 - 10x + 25 &= 9 \\
2x^2 - 14x + 20 &= 0 \\
x^2 - 7x + 10 &= 0 \\
(x-2)(x-5) &= 0
\end{aligned}$$

so the intersection points are $P(2, 1)$ and $Q(5, 4)$, as marked on the diagram.

Now $|w|$ is the distance from $O$ to the point representing $w$. From the diagram, the point of the region furthest from $O$ is $Q$, so the maximum value of $|w|$ is

$$|5 + 4i| = \sqrt{5^2 + 4^2} = \sqrt{41}$$

**3.** On an Argand diagram, the locus of points satisfying $|z-4+3i|=8$ is a circle.

**(a)** Write down

(i) the coordinates of the centre of this circle **[1]**

(ii) the radius of this circle **[1]**

**(b)** On an Argand diagram, shade the set of points defined by

$$|z-4+3i| \leq 8$$ **[1]**

**(c)** For the set of points defined in part (b), determine the maximum value of $|z|$. **[3]**

The set of points $A$ is defined by

$$A = \left\{ z \in \mathbb{C} \;\middle|\; -\frac{\pi}{4} \leq \arg(z-3-6i) \leq \frac{3\pi}{4} \right\} \bigcap \left\{ z \in \mathbb{C} \;\middle|\; |z-4+3i| \leq 8 \right\}$$

**(d)** Determine the area of the region defined by $A$, giving your answer to 3 significant figures. **[4]**

**Solution**

**(a)** (i)

$$|z-4+3i| = |z-(4-3i)|$$

So the circle has centre $(4, -3)$.

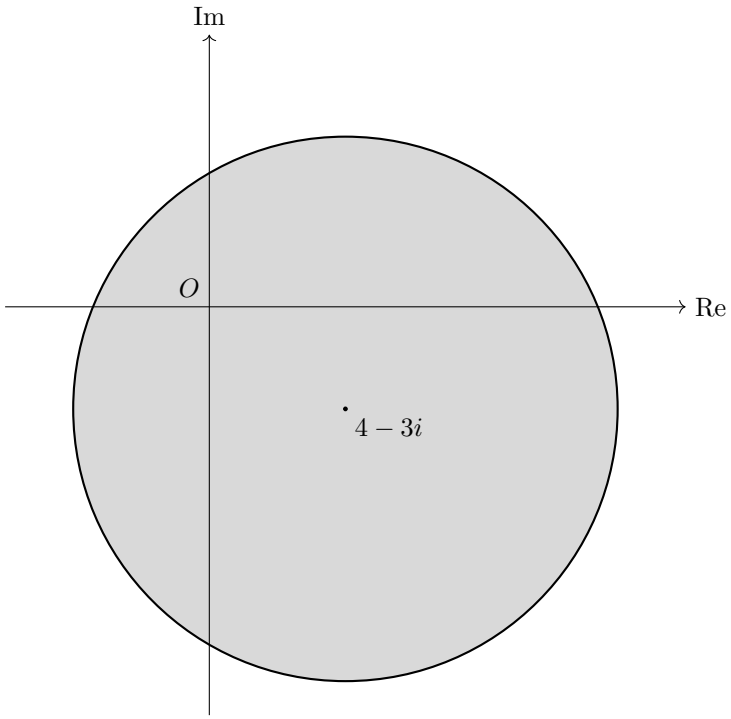
(ii) Comparing with the standard form $|z-a|=r$, the radius is 8.

**(b)** The inequality

$$|z-4+3i| \leq 8$$

means all points whose distance from $4-3i$ is at most 8.

So we shade the interior and boundary of the circle with centre $(4, -3)$ and radius 8.




**(c)** The modulus $|z|$ is the distance from the origin $O$ to the point representing $z$.

The centre $C$ of the circle represents $4 - 3i$, so

$$OC = |4 - 3i| = \sqrt{4^2 + (-3)^2} = 5$$

The point of the region furthest from $O$ is where the line from $O$ through $C$ meets the circle on the far side of $C$. Its distance from $O$ is $OC$ plus the radius.

Therefore the maximum value of $|z|$ is

$$5 + 8 = 13$$

**(d)** Let $z = x + iy$.

First consider the argument condition:

$$-\frac{\pi}{4} \leq \arg(z - 3 - 6i) \leq \frac{3\pi}{4}$$

Since

$$z - 3 - 6i = (x - 3) + (y - 6)i$$

the boundary arguments $-\frac{\pi}{4}$ and $\frac{3\pi}{4}$ are opposite directions, so they form a straight line through $(3, 6)$ with gradient $-1$.

Hence the boundary line is

$$y - 6 = -(x - 3)$$

so

$$x + y = 9$$

To decide which side is needed, take the point $(3, 7)$. Then

$$\arg\big((3+7i)-3-6i\big) = \arg(i) = \frac{\pi}{2}$$

which is in the given range, and $(3, 7)$ satisfies $x + y = 10$.

So the argument condition is the half-plane

$$x + y \geq 9$$

The circle has centre $C = (4, -3)$ and radius 8. Since

$$4 + (-3) = 1 < 9$$

the centre lies on the opposite side of the line, so the region $A$ is the minor segment cut off by the line $x + y = 9$.

The circle has equation $(x-4)^2 + (y+3)^2 = 64$. To find where the line meets it, substitute $y = 9 - x$:

$$(x-4)^2 + (12-x)^2 = 64$$

$$x^2 - 8x + 16 + x^2 - 24x + 144 = 64$$

$$2x^2 - 32x + 96 = 0$$

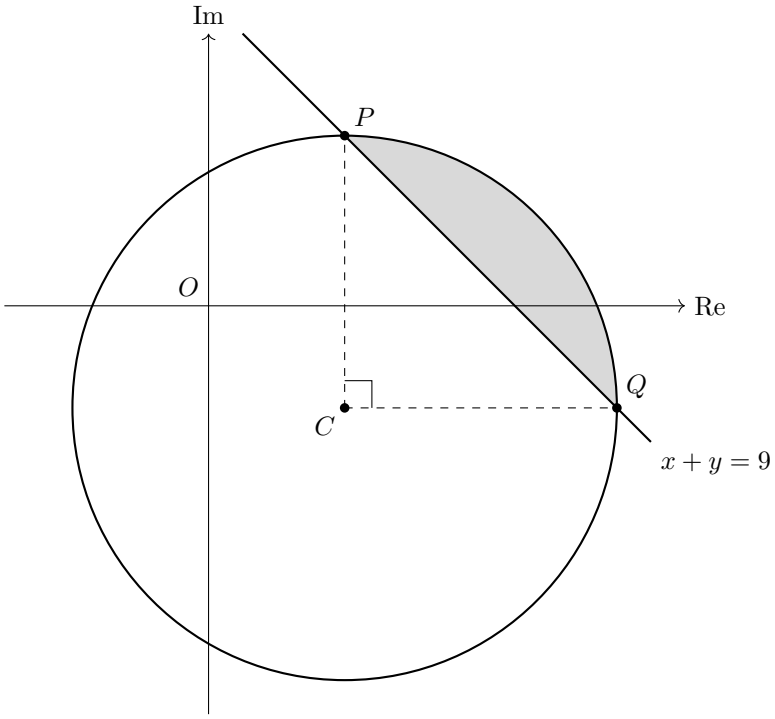
$$x^2 - 16x + 48 = 0$$

$$(x-4)(x-12) = 0$$

So the line meets the circle at $P = (4, 5)$ and $Q = (12, -3)$.

$CP$ joins $(4, -3)$ to $(4, 5)$, so it is vertical, and $CQ$ joins $(4, -3)$ to $(12, -3)$, so it is horizontal. Hence

$$\angle PCQ = \frac{\pi}{2}$$

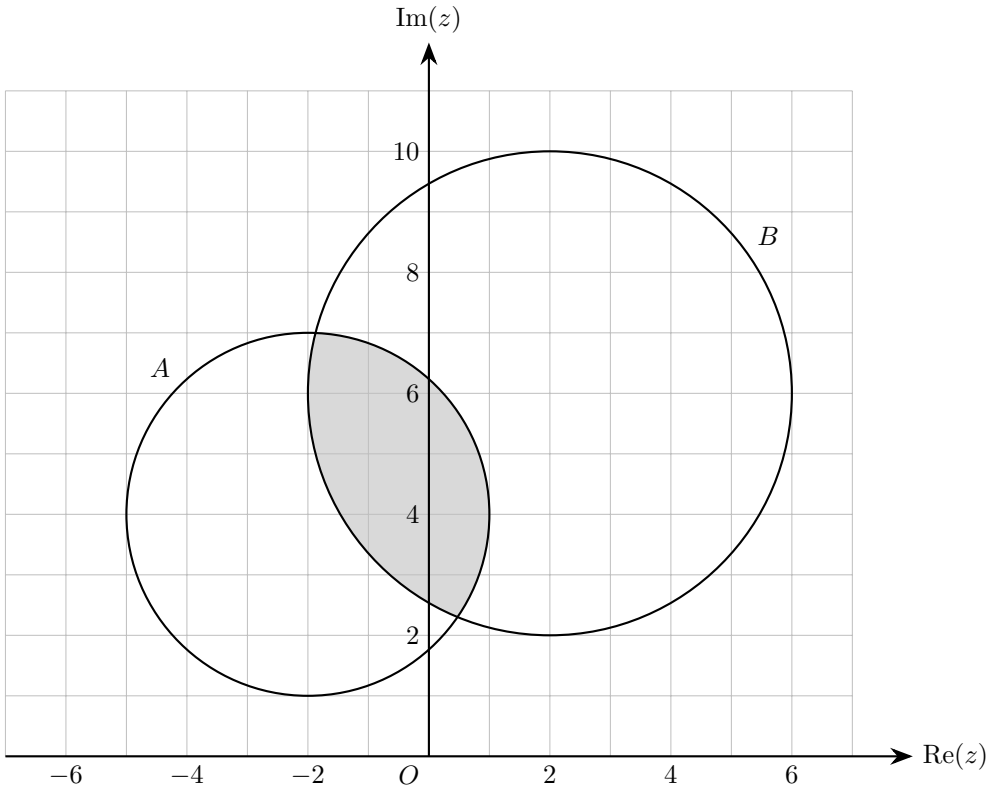


Therefore

$$\begin{aligned}\text{Area}(A) &= \text{area of sector } PCQ - \text{area of } \triangle PCQ \\ &= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta \\ &= \frac{1}{2}\times 8^2 \times \frac{\pi}{2} - \frac{1}{2}\times 8^2 \times \sin\frac{\pi}{2} \\ &= 16\pi - 32\end{aligned}$$

So the area is

$$16\pi - 32 \approx 18.3 \qquad (3 \text{ s.f.})$$




**4.** **(a)** The diagram above shows an Argand diagram.

(i) Write down the equation of the locus represented by the circumference of circle $A$. **[2]**

(ii) Write down the two inequalities that define the shaded region, not including either circumference. **[3]**

**(b)** (i) Draw an Argand diagram to show the region where

$$-\frac{\pi}{6} < \arg(z+2-i) < \frac{\pi}{4}$$ **[3]**

(ii) Determine whether the point $1+5i$ lies within this region. **[3]**

**Solution**

**(a)** (i) For a circle with centre $a$ and radius $r$, the locus is

$$|z-a| = r$$

Circle $A$ has centre $-2+4i$ and radius 3, so

$$|z-(-2+4i)| = 3$$

hence the circumference of circle $A$ is

$$|z+2-4i| = 3$$

(ii) The shaded region is the part inside both circles, and neither circumference is included.

Inside circle $A$:

$$|z+2-4i| < 3$$

Circle $B$ has centre $2+6i$ and radius 4, so inside circle $B$:

$$|z-(2+6i)|<4$$

So the two inequalities are

$$|z+2-4i|<3 \qquad \text{and} \qquad |z-(2+6i)|<4$$

**(b)** (i) We have

$$z+2-i=z-(-2+i)$$

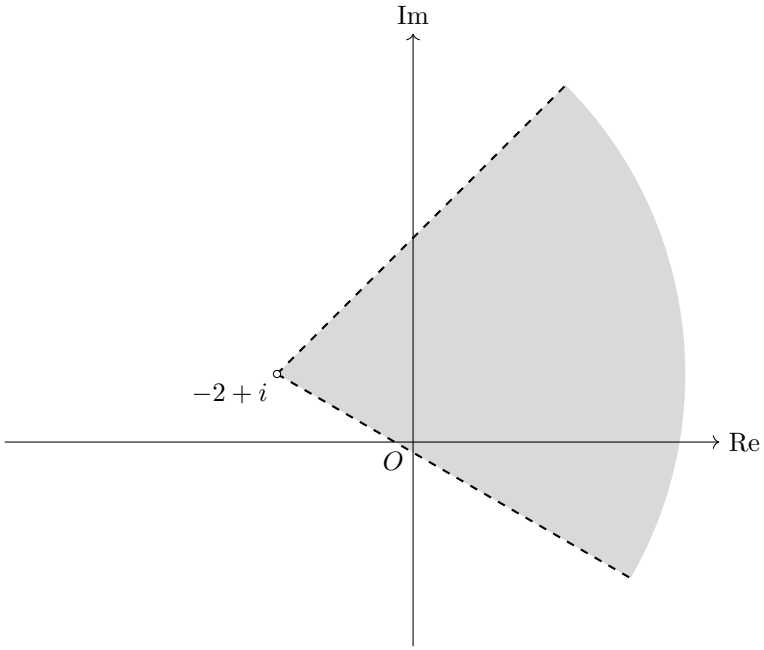
so the argument is measured from the point $-2+i$.

The region

$$-\frac{\pi}{6}<\arg(z+2-i)<\frac{\pi}{4}$$

is therefore the sector with vertex at $-2+i$ between the half-lines making angles $-\frac{\pi}{6}$ and $\frac{\pi}{4}$ with the positive real axis.

Since the inequalities are strict, both boundary half-lines are excluded. The vertex $-2+i$ is excluded too, as $\arg 0$ is undefined.




(ii) For the point $z=1+5i$:

$$z+2-i=(1+5i)+2-i=3+4i$$

$3+4i$ lies in the first quadrant, so

$$\arg(z+2-i)=\arg(3+4i)=\arctan\left(\frac{4}{3}\right)\approx 0.927$$

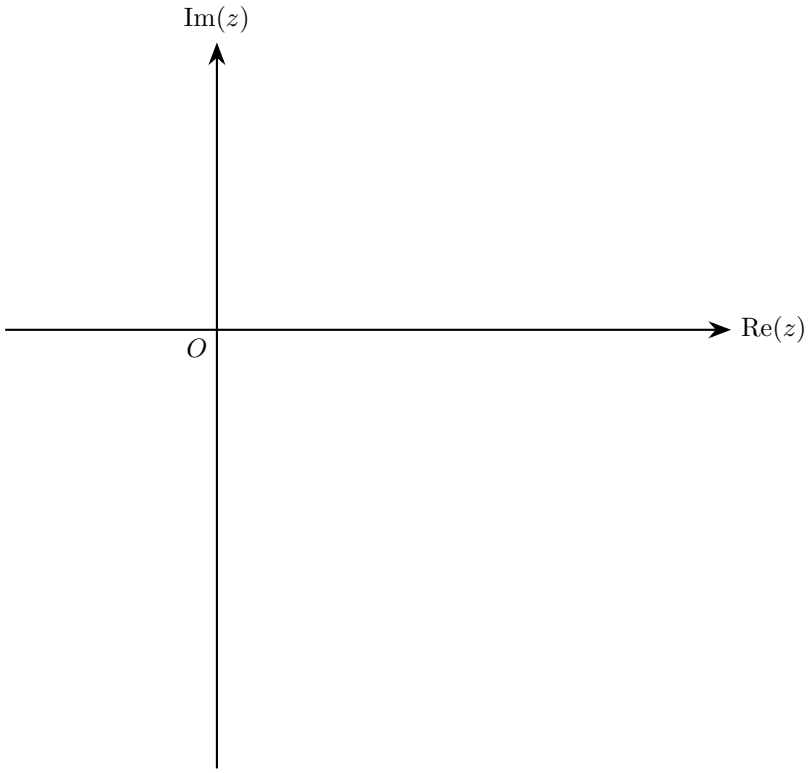
But

$$\frac{\pi}{4}\approx 0.785$$

Since

$$0.927>\frac{\pi}{4}$$

the argument is too large, so $1 + 5i$ does not lie in the region.




**5.** **(a)** Find the modulus of the complex number $2 + 2\sqrt{3}i$, giving your answer as an integer. **[2]**

**(b)** The locus of points, $L$, satisfies the equation $|z - (2 - 2\sqrt{3}i)| = 2$.

(i) Sketch the locus $L$ on the Argand diagram above. **[2]**

(ii) The complex number $w$ lies on $L$ so that $-\pi < \arg w \leq \pi$. Find the greatest possible value of $\arg w$, giving your answer in terms of $\pi$. **[2]**

**(c)** Solve the equation $z^4 = 2+2\sqrt{3}i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[5]**

**Solution**

**(a)** The modulus is

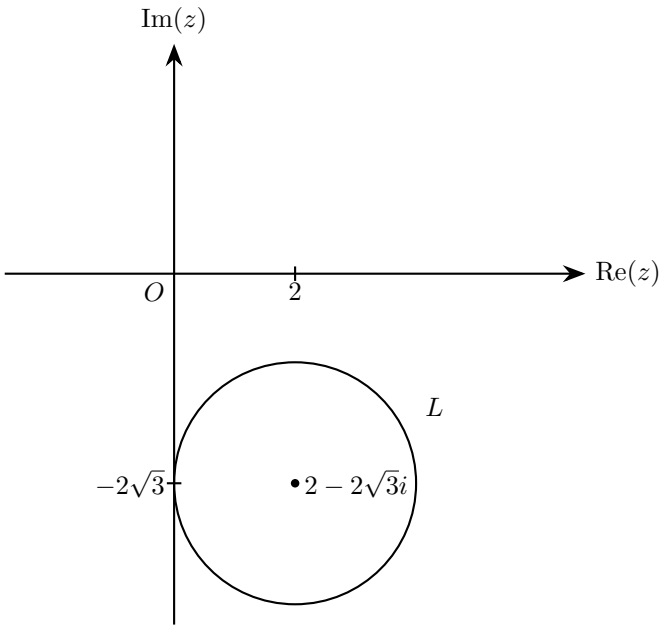
$$|2 + 2\sqrt{3}i| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$$

**(b)** (i) The equation

$$|z - (2 - 2\sqrt{3}i)| = 2$$

represents all points a distance 2 from the point $2 - 2\sqrt{3}i$.

So $L$ is a circle with centre $(2, -2\sqrt{3})$ and radius 2. The centre is a distance 2 from the imaginary axis, so the circle touches the imaginary axis at $-2\sqrt{3}i$.




(ii) Let $C$ be the centre of the circle, so

$$C = 2 - 2\sqrt{3}i$$

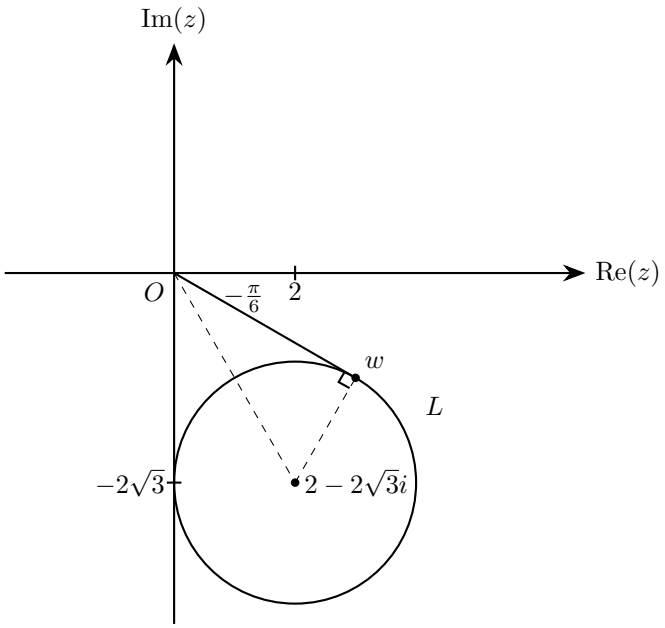
First find $OC$:

$$OC = |2 - 2\sqrt{3}i| = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$C$ lies in the fourth quadrant, so

$$\arg C = -\arctan \frac{2\sqrt{3}}{2} = -\arctan \sqrt{3} = -\frac{\pi}{3}$$

The greatest possible value of $\arg w$ occurs when the line from the origin to the circle is a tangent, touching the circle at $w$.




Then triangle $OCW$ is right-angled at $W$, with

$$CW = 2, \qquad OC = 4$$

Let $\angle COW = \alpha$. Then

$$\sin\alpha = \frac{CW}{OC} = \frac{2}{4} = \frac{1}{2}$$

so

$$\alpha = \frac{\pi}{6}$$

This tangent is above the line $OC$, so

$$\arg w = \arg C + \alpha = -\frac{\pi}{3} + \frac{\pi}{6} = -\frac{\pi}{6}$$

**(c)** From part (a):

$$|2 + 2\sqrt{3}i| = 4$$

and $2 + 2\sqrt{3}i$ lies in the first quadrant, so

$$\arg(2 + 2\sqrt{3}i) = \arctan\frac{2\sqrt{3}}{2} = \arctan\sqrt{3} = \frac{\pi}{3}$$

So

$$2 + 2\sqrt{3}i = 4e^{i\pi/3}$$

We solve

$$z^4 = 4e^{i\pi/3}$$

Let $z = re^{i\theta}$. Then

$$z^4 = r^4 e^{i4\theta}$$

so

$$r^4 e^{i4\theta} = 4e^{i(\pi/3+2k\pi)} \qquad (k \in \mathbb{Z})$$

Equating moduli gives

$$r^4 = 4$$

and $r > 0$, so

$$r = 4^{1/4} = \sqrt{2}$$

Equating arguments gives

$$4\theta = \frac{\pi}{3} + 2k\pi$$

hence

$$\theta = \frac{\pi}{12} + \frac{k\pi}{2}$$

The four distinct roots come from $k = 0, 1, 2, 3$:

$$\theta = \frac{\pi}{12}, \quad \frac{7\pi}{12}, \quad \frac{13\pi}{12}, \quad \frac{19\pi}{12}$$

Writing these in the range $-\pi < \theta \leq \pi$:

$$\theta = \frac{\pi}{12}, \quad \frac{7\pi}{12}, \quad -\frac{11\pi}{12}, \quad -\frac{5\pi}{12}$$

Therefore

$$z = \sqrt{2}e^{i\pi/12}, \qquad \sqrt{2}e^{i7\pi/12}, \qquad \sqrt{2}e^{-i11\pi/12}, \qquad \sqrt{2}e^{-i5\pi/12}$$

**6.** The point $P$ represents a complex number $z$ on an Argand diagram such that

$$|z-(1+i)| = \frac{1}{2}|z-(4+i)|$$

**(a)** Show that, as $z$ varies, the locus of $P$ is a circle, and give the coordinates of the centre and the radius of the circle. **[5]**

The point $Q$ represents a complex number $z$ on an Argand diagram such that

$$|z-(2+3i)| = |z+2-i|$$

**(b)** Sketch, on the same Argand diagram, the locus of $P$ and the locus of $Q$ as $z$ varies. **[5]**

**(c)** On your diagram shade the region which satisfies

$$|z-(1+i)| \leq \frac{1}{2}|z-(4+i)| \quad \text{and} \quad |z-(2+3i)| \geq |z+2-i|$$

**[2]**

**Solution**

**(a)** Let

$$z = x + iy$$

Then

$$|z-(1+i)| = \sqrt{(x-1)^2+(y-1)^2}$$

and

$$|z-(4+i)| = \sqrt{(x-4)^2+(y-1)^2}$$

So the given condition becomes

$$\sqrt{(x-1)^2+(y-1)^2} = \frac{1}{2}\sqrt{(x-4)^2+(y-1)^2}$$

Squaring both sides:

$$4\big((x-1)^2+(y-1)^2\big) = (x-4)^2+(y-1)^2$$

Expand:

$$4\big(x^2-2x+1+y^2-2y+1\big) = x^2-8x+16+y^2-2y+1$$

$$4x^2-8x+4+4y^2-8y+4 = x^2-8x+16+y^2-2y+1$$

Bring everything to one side:

$$3x^2+3y^2-6y-9 = 0$$

$$x^2+y^2-2y-3 = 0$$

Complete the square in $y$:

$$x^2 + (y^2 - 2y + 1) - 4 = 0$$
$$x^2 + (y-1)^2 = 4$$

This is the equation of a circle.

Hence the locus is the circle with centre $(0, 1)$ and radius 2.

**(b)** For the second locus, let again $z = x + iy$.

Then

$$|z - (2 + 3i)| = \sqrt{(x-2)^2 + (y-3)^2}$$

and

$$|z + 2 - i| = |z - (-2 + i)| = \sqrt{(x+2)^2 + (y-1)^2}$$

So

$$\sqrt{(x-2)^2 + (y-3)^2} = \sqrt{(x+2)^2 + (y-1)^2}$$

Squaring both sides:

$$(x-2)^2 + (y-3)^2 = (x+2)^2 + (y-1)^2$$

Expand and simplify:

$$x^2 - 4x + 4 + y^2 - 6y + 9 = x^2 + 4x + 4 + y^2 - 2y + 1$$
$$-4x - 6y + 13 = 4x - 2y + 5$$
$$-8x - 4y + 8 = 0$$
$$2x + y - 2 = 0$$

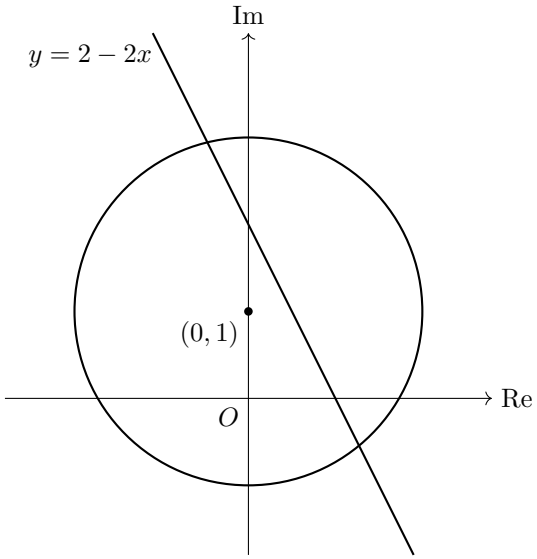
So the locus of $Q$ is the straight line

$$y = 2 - 2x$$

This is the perpendicular bisector of the line segment joining $(2, 3)$ and $(-2, 1)$.

On the same Argand diagram, we sketch:

- the circle $x^2 + (y-1)^2 = 4$, centre $(0, 1)$, radius 2,
- the line $y = 2 - 2x$.




**(c)** As in part (a), the inequality

$$|z - (1 + i)| \leq \frac{1}{2}|z - (4 + i)|$$

gives

$$x^2 + (y - 1)^2 \leq 4$$

so the point lies inside or on the circle.

As in part (b), the inequality

$$|z - (2 + 3i)| \geq |z + 2 - i|$$

gives

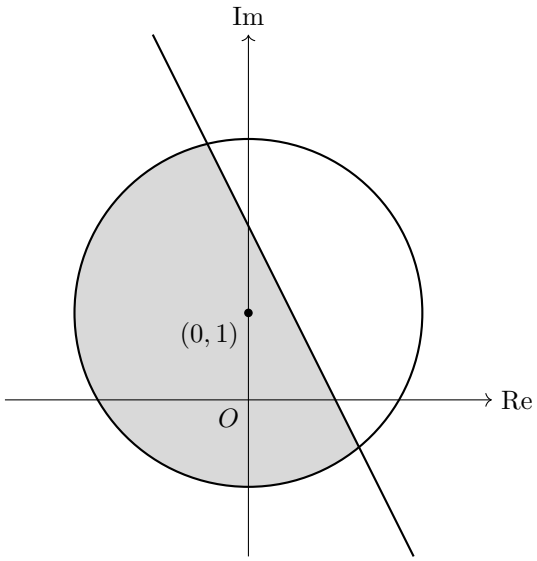
$$\begin{aligned}(x - 2)^2 + (y - 3)^2 &\geq (x + 2)^2 + (y - 1)^2 \\ -8x - 4y + 8 &\geq 0 \\ 2x + y - 2 &\leq 0 \\ y &\leq 2 - 2x\end{aligned}$$

So the point lies on or below the line $y = 2 - 2x$.

Therefore the required region is the intersection of:

- the inside of the circle $x^2 + (y - 1)^2 = 4$,
- the half-plane $y \leq 2 - 2x$.

The shaded region is shown below.



**7.** On an Argand diagram, the point $P$ representing the complex number $z$ lies on the locus

$$|z-(2-i)|=3$$

The point $Q$ representing the complex number $w$ lies on the locus

$$|w+1-11i|=3|w+1-3i|$$

Find the exact coordinates of the points common to these two loci. **[7]**

**Solution**

Let a point common to both loci represent the complex number

$$u=x+iy$$

so its coordinates are $(x,y)$. (A new letter is used because the question uses $z$ and $w$ for the two loci.)

For the locus of $P$:

$$|u-(2-i)|=3$$

Now

$$u-(2-i)=(x-2)+(y+1)i$$

so

$$\sqrt{(x-2)^2+(y+1)^2}=3$$

Squaring both sides:

$$(x-2)^2+(y+1)^2=9$$

For the locus of $Q$:

$$|u+1-11i|=3|u+1-3i|$$

Now

$$u+1-11i=(x+1)+(y-11)i, \qquad u+1-3i=(x+1)+(y-3)i$$

So

$$\sqrt{(x+1)^2+(y-11)^2}=3\sqrt{(x+1)^2+(y-3)^2}$$

Squaring both sides:

$$(x+1)^2+(y-11)^2=9\big((x+1)^2+(y-3)^2\big)$$

Expand and simplify:

$$\begin{aligned} x^2+2x+1+y^2-22y+121&=9x^2+18x+9+9y^2-54y+81\\ 0&=8x^2+16x+8+8y^2-32y-40\\ 0&=x^2+2x+y^2-4y-4\\ 0&=(x+1)^2+(y-2)^2-9 \end{aligned}$$

Hence the second locus is

$$(x+1)^2 + (y-2)^2 = 9$$

So the common points satisfy the simultaneous equations

$$(x-2)^2 + (y+1)^2 = 9$$

and

$$(x+1)^2 + (y-2)^2 = 9$$

Both right-hand sides equal 9, so equating the left-hand sides:

$$(x-2)^2 + (y+1)^2 = (x+1)^2 + (y-2)^2$$

$$x^2 - 4x + 4 + y^2 + 2y + 1 = x^2 + 2x + 1 + y^2 - 4y + 4$$

$$-6x + 6y = 0$$

$$y = x$$

Substitute $y = x$ into $(x-2)^2 + (y+1)^2 = 9$:

$$(x-2)^2 + (x+1)^2 = 9$$

$$x^2 - 4x + 4 + x^2 + 2x + 1 = 9$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

So

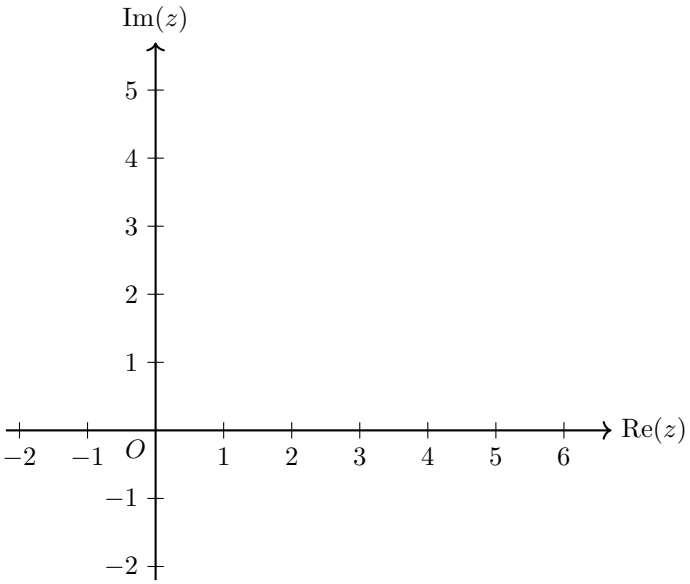
$$x = 2 \quad \text{or} \quad x = -1$$

and $y = x$, so

$$y = 2 \quad \text{or} \quad y = -1$$

Therefore the exact coordinates of the common points are

$$(2, 2) \quad \text{and} \quad (-1, -1)$$




**8.** **(a)** Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z-(3+i)|=1$$ **[2]**

**(b)** There is a unique complex number $w$ that satisfies both

$$|w-(3+i)|=1 \quad \text{and} \quad \arg(w)=\alpha$$

where $\alpha$ is a constant such that $0<\alpha<\pi$.

(i) Find the value of $\alpha$. **[2]**

(ii) Find $w$ and express it in the form $r(\cos\theta+i\sin\theta)$. Give each of $r$ and $\theta$ to 2 significant figures. **[4]**

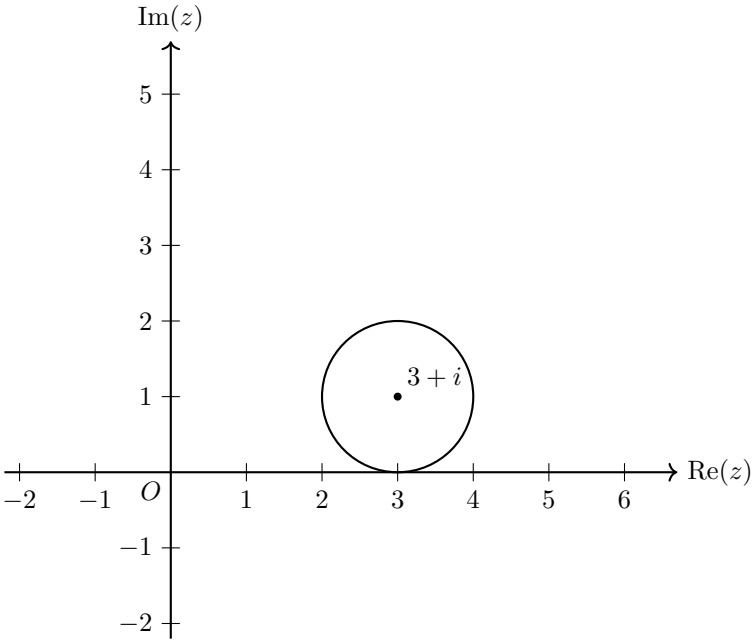
**Solution**

**(a)** The locus

$$|z-(3+i)|=1$$

is the set of points whose distance from $3+i$ is 1.

So it is a circle with centre $(3, 1)$ and radius 1.




**(b)** (i) If $\arg(w) = \alpha$, then $w$ lies on the half-line from the origin at angle $\alpha$. This half-line is part of the line

$$y = mx$$

where

$$m = \tan\alpha$$

For there to be a unique point $w$ satisfying both conditions, the half-line must touch the circle at exactly one point, so it must be a tangent.

The circle has equation $(x-3)^2 + (y-1)^2 = 1$. Substituting $y = mx$:

$$(x-3)^2 + (mx-1)^2 = 1$$

$$x^2 - 6x + 9 + m^2x^2 - 2mx + 1 = 1$$

$$(1+m^2)x^2 - (6+2m)x + 9 = 0$$

For a tangent this quadratic has a repeated root, so its discriminant is zero:

$$(6+2m)^2 - 4(1+m^2)(9) = 0$$

$$36 + 24m + 4m^2 - 36 - 36m^2 = 0$$

$$24m - 32m^2 = 0$$

$$8m(3-4m) = 0$$

So

$$m = 0 \quad \text{or} \quad m = \frac{3}{4}$$

If $m = 0$, then $\alpha = 0$ (the real axis, which touches the circle at 3), but the question states $0 < \alpha < \pi$, so this is not allowed.

Hence

$$\tan\alpha = \frac{3}{4}$$

so

$$\alpha = \arctan\left(\frac{3}{4}\right) \approx 0.644$$

(ii) Using the value of $m$, the tangent line is

$$y = \frac{3}{4}x$$

The circle equation is

$$(x-3)^2 + (y-1)^2 = 1$$

Substituting $y = \frac{3}{4}x$:

$$(x-3)^2 + \left(\frac{3}{4}x - 1\right)^2 = 1$$

$$x^2 - 6x + 9 + \frac{9}{16}x^2 - \frac{3}{2}x + 1 = 1$$

$$\frac{25}{16}x^2 - \frac{15}{2}x + 9 = 0$$

Multiplying by 16:

$$25x^2 - 120x + 144 = 0$$

Factorising:

$$(5x - 12)^2 = 0$$

So

$$x = \frac{12}{5}$$

Then

$$y = \frac{3}{4} \times \frac{12}{5} = \frac{9}{5}$$

Hence

$$w = \frac{12}{5} + \frac{9}{5}i$$

Now find its modulus:

$$|w| = \sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{9}{5}\right)^2}$$

$$= \sqrt{\frac{144}{25} + \frac{81}{25}}$$

$$= \sqrt{\frac{225}{25}}$$

$$= 3$$

Its argument is

$$\theta = \arg(w) = \alpha = \arctan\left(\frac{3}{4}\right) \approx 0.64$$

So

$$w = \frac{12}{5} + \frac{9}{5}i = 3.0(\cos 0.64 + i \sin 0.64) \qquad (2 \text{ s.f.})$$

**9.** In an Argand diagram, the point $P$ representing the complex number $w$ lies on the locus defined by

$$\left\{z \in \mathbb{C} \;\middle|\; \arg(z-2+i) = \frac{\pi}{6}\right\}$$

You are given that $\text{Im}(w) = 3$.

**(a)** Find $w$. **[2]**

The point $P$ also lies on the locus defined by $\left\{z \in \mathbb{C} \;\middle|\; |z-2-11i| = k\right\}$, where $k$ is a constant.

**(b)** Find the complex number represented by the other point of intersection of the loci defined by

$$\left\{z \in \mathbb{C} \;\middle|\; |z-2-11i| = k\right\} \quad \text{and} \quad \left\{z \in \mathbb{C} \;\middle|\; \arg(z-2+i) = \frac{\pi}{6}\right\}$$

**[7]**

**Solution**

**(a)** Let

$$w = x + 3i$$

since $\text{Im}(w) = 3$.

Then

$$w - 2 + i = (x-2) + 4i$$

Given

$$\arg(w-2+i) = \frac{\pi}{6}$$

so

$$\tan\frac{\pi}{6} = \frac{4}{x-2}$$

Hence

$$\frac{1}{\sqrt{3}} = \frac{4}{x-2}$$

so

$$x - 2 = 4\sqrt{3}$$

and therefore

$$x = 2 + 4\sqrt{3}$$

Thus

$$w = 2 + 4\sqrt{3} + 3i$$

**(b)** From part (a):

$$P \text{ represents } w = 2 + 4\sqrt{3} + 3i$$

The second locus is

$$|z-2-11i| = k = |z-(2+11i)|$$

which is a circle with centre $2+11i$.

Using the point $P$ to find $k$:

$$\begin{aligned} k &= |w-(2+11i)| \\ &= \left|(2+4\sqrt{3}+3i)-(2+11i)\right| \\ &= \left|4\sqrt{3}-8i\right| \end{aligned}$$

So

$$\begin{aligned} k^2 &= (4\sqrt{3})^2+(-8)^2 \\ &= 48+64 \\ &= 112 \end{aligned}$$

and hence

$$k = 4\sqrt{7}$$

Now let a general point of intersection be

$$z = x+iy$$

Since

$$\arg(z-2+i) = \frac{\pi}{6}$$

the point lies on the half-line starting at $(2,-1)$ with gradient

$$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

so its equation is

$$y+1 = \frac{x-2}{\sqrt{3}}$$

that is

$$y = \frac{x-2}{\sqrt{3}} - 1$$

The circle has equation

$$(x-2)^2+(y-11)^2 = 112$$

Substitute $y = \frac{x-2}{\sqrt{3}} - 1$:

$$(x-2)^2 + \left(\frac{x-2}{\sqrt{3}} - 12\right)^2 = 112$$

Let

$$u = x-2$$

Then

$$u^2 + \left(\frac{u}{\sqrt{3}} - 12\right)^2 = 112$$

Expanding:

$$u^2 + \left(\frac{u^2}{3} - 8\sqrt{3}\,u + 144\right) = 112$$

$$\frac{4u^2}{3} - 8\sqrt{3}\,u + 144 = 112$$

$$\frac{4u^2}{3} - 8\sqrt{3}\,u + 32 = 0$$

Multiply by 3:

$$4u^2 - 24\sqrt{3}\,u + 96 = 0$$

Divide by 4:

$$u^2 - 6\sqrt{3}\,u + 24 = 0$$

Factorising:

$$(u - 4\sqrt{3})(u - 2\sqrt{3}) = 0$$

So

$$u = 4\sqrt{3} \quad \text{or} \quad u = 2\sqrt{3}$$

Both values are positive, so both points have $x > 2$ and lie on the half-line itself, not on its extension. The value $u = 4\sqrt{3}$ gives the known point $P$ from part (a), so the other intersection has

$$u = 2\sqrt{3}$$

Therefore

$$x = 2 + 2\sqrt{3}$$

and

$$y = \frac{2\sqrt{3}}{\sqrt{3}} - 1 = 2 - 1 = 1$$

Hence the other point represents

$$z = 2 + 2\sqrt{3} + i$$

**10.** The locus of points $z = x + iy$ that satisfy

$$\arg\left(\frac{z-(2+i)}{z-(6+5i)}\right) = \frac{\pi}{4}$$

is an arc of a circle $C$.

**(a)** On an Argand diagram sketch the locus of $z$. **[2]**

**(b)** Explain why the centre of $C$ lies on the line $y = 7 - x$. **[1]**

**(c)** Determine the radius of $C$. **[2]**

**(d)** Determine the coordinates of the centre of $C$. **[2]**

**Solution**

**(a)** Let

$$A = 2 + i \quad \text{and} \quad B = 6 + 5i$$

so $A = (2, 1)$ and $B = (6, 5)$.

The condition

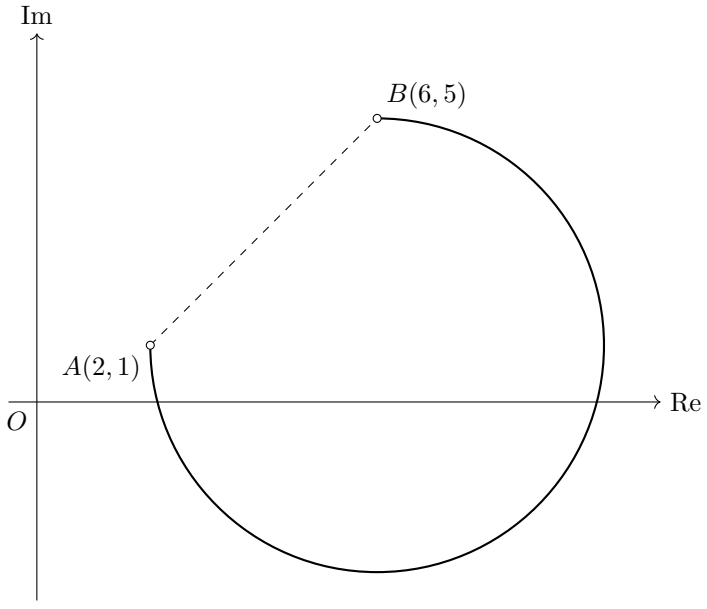
$$\arg\left(\frac{z-A}{z-B}\right) = \frac{\pi}{4}$$

means

$$\arg(z-A) - \arg(z-B) = \frac{\pi}{4}$$

so, if $Z$ is the point representing $z$, the angle $AZB$ is $\frac{\pi}{4}$.

Angles in the same segment are equal, so the locus is an arc of a circle through $A$ and $B$. Since $\frac{\pi}{4} < \frac{\pi}{2}$, it is the major arc. It lies below and to the right of the line segment $AB$, as checked in part (d). Also, $A$ and $B$ are excluded because the expression is undefined there.



**(b)** Since $A$ and $B$ lie on circle $C$, the centre of $C$ is equidistant from $A$ and $B$. Therefore it lies on the perpendicular bisector of $AB$.

The midpoint of $AB$ is

$$\left(\frac{2+6}{2}, \frac{1+5}{2}\right) = (4, 3)$$

The gradient of $AB$ is

$$\frac{5-1}{6-2} = 1$$

so the perpendicular bisector has gradient $-1$.

Its equation is

$$y - 3 = -1(x - 4)$$

$$y = 7 - x$$

Therefore the centre lies on the line $y = 7 - x$.

**(c)** Let $P$ be the centre of $C$ and $r$ its radius.

The angle at the circumference is $\frac{\pi}{4}$, so the angle at the centre subtended by the same chord $AB$ is

$$\angle APB = 2 \times \frac{\pi}{4} = \frac{\pi}{2}$$

So triangle $APB$ is right-angled at $P$, with $PA = PB = r$. Also

$$AB^2 = (6-2)^2 + (5-1)^2 = 32$$

By Pythagoras:

$$r^2 + r^2 = 32$$

so $r^2 = 16$, giving

$$r = 4$$

**(d)** From part (b), the centre $P$ lies on $y = 7 - x$, so write $P = (a,\ 7 - a)$.

From part (c), $PA = 4$, so

$$(a-2)^2 + (7-a-1)^2 = 16$$

$$a^2 - 4a + 4 + a^2 - 12a + 36 = 16$$

$$2a^2 - 16a + 24 = 0$$

$$a^2 - 8a + 12 = 0$$

$$(a-2)(a-6) = 0$$

Hence the two possible centres are $(2, 5)$ and $(6, 1)$.

To decide which is correct, test a point on one of the two circles. The point $z = 10 + i$ lies on the circle centred at $(6, 1)$, diametrically opposite $A$:

$$\begin{aligned}\arg\left(\frac{(10+i)-(2+i)}{(10+i)-(6+5i)}\right) &= \arg\left(\frac{8}{4-4i}\right)\\ &= \arg\left(\frac{8(4+4i)}{(4-4i)(4+4i)}\right)\\ &= \arg(1+i)\\ &= \frac{\pi}{4}\end{aligned}$$

So $10 + i$ lies on the locus. But

$$\text{distance from } (10, 1) \text{ to } (2, 5) = \sqrt{(8)^2 + (-4)^2} = \sqrt{80} \neq 4$$

so it does not lie on the circle centred at $(2, 5)$. This also confirms that the arc lies below and to the right of $AB$.

Therefore the centre of $C$ is $(6, 1)$.